

APP Layer — Part A Test Sequence

ALL STEPS PASSED

Modules: LIGHT · DOOR · CLIMATE · HEATER · ALARM · SEC

Firmware test_app (branch phase2-app) · Proteus run 2026-10-04 · Tester: Abdelrahman Sherif Medhat

9	48	6	0
test blocks	checked steps	APP modules	failures

This sequence verifies the six APP modules that work without a user interface. Typed terminal lines stand in for the LCD/keypad and UART menus, which arrive in part B. Run the blocks in order: the lockdown block is last because only an MCU RESET ends it. Step IDs in brackets refer to **docs/test_plan.md** §3.

0 Setup

Part	Setting
ATmega32	Program File = .pio/build/test_app/firmware.hex · Clock Frequency = 16 MHz (must equal F_CPU)
Virtual Terminal	9600 baud, 8N1 · “Echo Typed Characters” unchecked (TERM echoes itself) · click inside before typing, end every line with Enter
LM35 start values	Ambient (PA0) = 25 °C · Water (PA1) = 40 °C
Oscilloscope	Channel A = PB3 (dimmer) · B = PD4 (AC fan) · C = PD5 (door servo)
I²C debugger	Optional, on SDA/SCL — used for the EEPROM write in step H-9
Servo	Min / Max Angle = 0 / 180 (the Proteus default maps 1-2 ms to −90...+90°)

Averaging rule. Temperature decisions use the average of the last 10 samples taken every 100 ms. After every LM35 change, wait about **2 s** before judging the result.

Terminal reference

Line / command	Meaning
[INFO] AC 1 / AC 0	AC switched on / off
[INFO] HEATER_POWER 1/0	heater panel on / off
[INFO] HEATER_ELEMENT n	0 = none, 1 = heating, 2 = cooling
[INFO] HEATER_SET n	new set temperature (panel or terminal)
[INFO] LAMP n · DIMMER n	lamp n toggled · new dimmer level in %
[STAT] room ... el ...	snapshot every 5 s: room temp, AC, water temp, set temp, heater on, element
1 ... 5	toggle lamp n
+ / -	dimmer up / down 10 %
o / c	open / close the door
35 ... 75	heater set temperature (other values: “[INFO] rejected”)
a	trigger the lockdown (ALARM)

1 Automatic part

About 2 s after Play. Do not touch any button: the test pulls the button pins low itself, so lamps, relays, servo, fan and buzzer twitch briefly.

#	Do	Expect	Result
A-1	Wait for the SUMMARY line	Every line is <code>[PASS]</code> ; SUMMARY reports 0 failed	PASS
A-2	Read the next line	<code>LIVE: heater panel + AC ...</code> appears; the live super-loop is running	PASS

2 Start-up state

[T-40, T-51]

#	Do	Expect	Result
S-1	Look at lamps, dimmer, door	All 5 lamps off · scope A flat low (0 %) · servo at 0°	PASS
S-2	Look at the heater panel and AC	7-segment blank · heater LED (PA3) off · red (PB6) and blue (PB7) off · fan stopped	PASS
S-3	Wait 5 s	<code>[STAT] room 25 ac 0 water 40 set 60 on 0 el 0</code> (set = 60 or the last stored value)	PASS

3 Lights and door

[T-41, T-42, T-44, T-45]

#	Do	Expect	Result
L-1	Type 1, 3, 5	Lamps 1, 3, 5 on; <code>[INFO] LAMP 1 / 3 / 5</code>	PASS
L-2	Type 3	Only lamp 3 turns off	PASS
L-3	Type + once	<code>[INFO] DIMMER 10</code> ; scope A ≈ 10 % duty at ≈ 7.8 kHz	PASS
L-4	Type + four more times	50 % duty; dimmer lamp brighter	PASS
L-5	Keep typing +	Stops at 100 % (flat high); more presses change nothing	PASS
L-6	Type - down to 0	0 % flat low; dimmer lamp off	PASS
D-1	Type o, then o	<code>[INFO] D00R moved 1</code> , servo to 90°; then <code>moved 0</code> , no movement	PASS
D-2	Type c, then c	<code>moved 1</code> , servo back to 0°; then <code>moved 0</code>	PASS
D-3	Scope C, closed vs open	Period 20 ms; pulse 1.0 ms closed, 1.5 ms open	PASS

4 Air conditioning

[T-46 – T-50] Change only the ambient LM35 (PA0).

#	Do	Expect	Result
C-1	Ambient 25 °C	<code>[STAT] ... room 25 ac 0</code> ; fan stopped	PASS
C-2	Ambient 28 °C, wait 2 s	Still OFF — the rule is strictly “above 28”	PASS
C-3	Ambient 29 °C	Within ≈ 1 s: <code>[INFO] AC 1</code> , fan runs, scope B shows PWM / high	PASS
C-4	Ambient 25, then 21 °C	Stays ON at both (hysteresis)	PASS
C-5	Ambient 20 °C	<code>[INFO] AC 0</code> , fan stops, scope B flat low	PASS
C-6	Ambient 25 °C	Stays OFF	PASS

5 Heater — power and setting mode

[T-52 – T-59, T-70, T-71]

#	Do	Expect	Result
H-1	Heater OFF: press Up, then Down	Nothing happens (decision D-10)	PASS
H-2	Press and hold ON/OFF for 2 s	Nothing while held	PASS
H-3	Release ON/OFF	[INFO] HEATER_POWER 1; display shows 40 (water temperature)	PASS
H-4	Press Up once	Display blinks 60 (0.5 s on / 0.5 s off); value not changed	PASS
H-5	Up, Up, Down	65, 70, 65 — each shown immediately	PASS
H-6	Keep pressing Up	Stops at 75	PASS
H-7	Keep pressing Down	Stops at 35	PASS
H-8	Press Up every 3 s, four times	Setting mode stays active throughout	PASS
H-9	Set 50 , then stop pressing	5 s later blinking stops, water temp shows; [INFO] HEATER_SET 50; I ² C: one page write to 0xA0, addr 0x10, data 0x32	PASS
H-10	Type 55	[INFO] HEATER_SET 55 — the remote path (HTR-14)	PASS
H-11	Type 62 , then 80	[INFO] rejected twice	PASS
H-12	Enter setting mode (Up), then type 45	Blinking value becomes 45 and the 5 s timer restarts	PASS

6 Heater — element control and LED

[T-63 – T-69] First type **60** and wait 5 s. Then change only the water LM35 (PA1).

#	Do	Expect	Result
E-1	Heater ON, water 40 → 50 in one step	Display climbs 40 → 50 over ≈ 1 s	PASS
E-2	Heater OFF, water 40; turn ON, watch red	Red on ≈ 1 s after turn-on, not instantly; HEATER_ELEMENT 1	PASS
E-3	While heating	Heater LED (PA3) blinks 0.5 s on / 0.5 s off	PASS
E-4	Water 50 → 54 → 56 → 64	Still heating at all four (red on, blue off)	PASS
E-5	Water 66	Red off, blue on, HEATER_ELEMENT 2; heater LED steady on	PASS
E-6	Water 62 → 56	Still cooling	PASS
E-7	Water 54	Blue off, red on, HEATER_ELEMENT 1; red and blue never on together	PASS
E-8	Press ON/OFF while heating	Red, blue, heater LED and display all off; HEATER_POWER 0	PASS
E-9	Heater OFF, water 70	Nothing switches on; [STAT] ... water 70 still updates	PASS

7 Persistence

[T-61]

#	Do	Expect	Result
P-1	Type 50 , wait 1 s, press RESET	Automatic part runs again, 0 failed	PASS
P-2	Heater ON, press Up once	Blinking 50 — the value came back from the EEPROM	PASS

8 Lockdown — last

[T-73, T-74] Prepare: heater heating (water 40, set 60) · AC on (ambient 29) · type 1, 2 · + ×5 · o.

#	Do	Expect	Result
K-1	Type a	[INFO] LOCKDOWN 0; buzzer beeps 0.5 s / 0.5 s	PASS
K-2	Check the forced outputs	Red and blue off · fan stopped · 7-segment blank · heater LED off	PASS
K-3	Check the kept outputs	Lamps 1, 2 still on · dimmer still 50 % · door still open	PASS
K-4	Press heater buttons; type 3, +, c	All ignored	PASS
K-5	Press RESET	Buzzer stops; normal start, 0 failed	PASS

Result — 2026-10-04. All blocks 0–8 executed in Proteus on the full schematic; every step matched its expectation. The APP part A modules (LIGHT, DOOR, CLIMATE, HEATER, ALARM, SEC) are verified. Next: part B (UILOC, UIREM) and the full test_plan.md §3 run.